QUADRATICS, REPEATING TAILS AND PELL ON ONE PAGE

Metallic ratios as continued fractions

An irrational number can contain a copy of its own rule. Unfold that rule to build rational approximations.

1 / AN INTEGER AND A LITTLE BIT

Isolate the bit, then substitute it back into itself

For integer $\lambda \geq 1$, write the positive root of $x^2 - \lambda x - 1 = 0$ as $\mu_\lambda = \lambda + t$, where $0 < t < 1$.

$$(\lambda + t)^2 - \lambda(\lambda + t) - 1 = 0 \quad \Longleftrightarrow \quad t^2 + \lambda t = 1 \quad \Longleftrightarrow \quad \boxed{t = \frac{1}{\lambda + t}}.$$

Replace each t by that same fraction, again and again:

$$\boxed{\mu_\lambda = \lambda + \cfrac{1}{\lambda + \cfrac{1}{\lambda + \ddots}} = [\overline{\lambda}] = \frac{\lambda + \sqrt{\lambda^2 + 4}}{2}.}$$

In $[a_0; a_1, a_2, \ldots]$, a_0 is the integer part; each later entry sits under the next reciprocal. The bar means *repeat this block for ever*.

2 / CHANGE THE INTEGER, KEEP THE CONSTRUCTION

The quadratic chooses the repeated entry

Stop at an integer entry and evaluate from the bottom. Each finite cutoff is a rational approximation called a **convergent**.

λ	ratio	quadratic	continued fraction	first four convergents
1	Golden	$x^2 - x - 1 = 0$	$[\overline{1}]$	$1,\ 2,\ 3/2,\ 5/3$
2	Silver	$x^2 - 2x - 1 = 0$	$[\overline{2}]$	$2,\ 5/2,\ 12/5,\ 29/12$
3	Bronze	$x^2 - 3x - 1 = 0$	$[\overline{3}]$	$3,\ 10/3,\ 33/10,\ 109/33$
4	4-metallic	$x^2 - 4x - 1 = 0$	$[\overline{4}]$	$4,\ 17/4,\ 72/17,\ 305/72$
5	5-metallic	$x^2 - 5x - 1 = 0$	$[\overline{5}]$	$5,\ 26/5,\ 135/26,\ 701/135$

3 / MAKE THE CONVERGENTS BY HAND

Two rows do the arithmetic

For $[a_0; a_1, a_2, \ldots]$, write the convergent as p_n/q_n. Multiply the last column by a_n, then add the column before it:

$$p_n = a_n p_{n-1} + p_{n-2},$$
$$q_n = a_n q_{n-1} + q_{n-2}.$$

For silver, every $a_n = 2$:

n	-2	-1	0	1	2	3	4
a_n			2	2	2	2	2
p_n	0	1	2	5	12	29	70
q_n	1	0	1	2	5	12	29

For example, $12 = 2{\cdot}5{+}2$ and $5 = 2{\cdot}2{+}1$. The shaded columns are starting values; $1/0$ is a formal starter, not a convergent.

4 / WHY UNFOLDING REALLY REACHES THE ROOT

Alternating errors get smaller

For a metallic ratio, put $C_0 = \lambda$, $C_{n+1} = \lambda + 1/C_n$, and $\mu = \mu_\lambda$. Subtract the root's rule, $\mu = \lambda + 1/\mu$:

$$C_{n+1} - \mu = \frac{1}{C_n} - \frac{1}{\mu} = -\frac{C_n - \mu}{\mu C_n}.$$

The sign flips, so approximations alternate below and above. Since $\mu C_n \geq \mu\lambda > 1$, each error shrinks by a factor at most $1/(\lambda\mu) < 1$. Therefore $C_n \to \mu$. For silver, later convergents give the bounds

$$\boxed{\frac{70}{29} < 1 + \sqrt{2} < \frac{29}{12}.}$$

Their gap is $29/12 - 70/29 = 1/348$.

5 / THE SILVER FRACTION MEETS PELL

Subtract one to approximate $\sqrt{2}$

Since $\mu_2 = 1 + \sqrt{2}$,

$$\sqrt{2} = [1; \overline{2}], \qquad \frac{p_n - q_n}{q_n} \longrightarrow \sqrt{2}.$$

Put $X_n = p_n - q_n$. The column rule gives

$$X_n^2 - 2q_n^2 = (-1)^{n+1}.$$

The first pair $(X_0, q_0) = (1, 1)$ gives -1; each new column reverses that sign.

X_n/q_n	$1/1$	$3/2$	$7/5$	$17/12$	$41/29$
$X_n^2 - 2q_n^2$	-1	1	-1	1	-1

For square triangular numbers, put $X = 2m + 1$ and $q = 2s$:

$$\frac{m(m+1)}{2} = s^2 \quad \Longleftrightarrow \quad X^2 - 8s^2 = 1.$$

The pair $(X, q) = (17, 12)$ gives $m = 8$, $s = 6$: $T_8 = 36 = 6^2$.

6 / THE SAME MOVE FOR ANOTHER QUADRATIC

What is the bit in $\sqrt{6} = 2 + t$?

Follow the same substitution:

$$(2 + t)^2 = 6 \implies t^2 + 4t = 2 \implies t = \frac{2}{4 + t}.$$

$$\boxed{\sqrt{6} = 2 + \cfrac{2}{4 + \cfrac{2}{4 + \ddots}}.}$$

This is a **generalised continued fraction**: the repeated numerators are 2. To obtain the usual numerators of 1, take reciprocals:

$$\frac{1}{t} = 2 + \frac{t}{2}, \qquad \frac{1}{t/2} = 4 + t.$$

The tail returns after $2, 4$, so $\sqrt{6} = [2; \overline{2, 4}]$.

Further reading.
J. B. Gil and A. Worley, *Generalized Metallic Means*, §1; K. Conrad, *Pell's Equation, II*, §5. Companions: metallic ratios and Binet forms; square triangular numbers.

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